No.	Status	Date
QSCP-CI-SUM-2008-01	Issued	2008-02-08

Lesson Learned - Localized Corrosion at Uncovered Elbows

Incident Summary

A leak occurred on 2-RV-102, a 60.3 mm OD A106 Gr.B regenerator overhead line in the ATU-100. The line has a nominal wall thickness of 5.54 mm and a minimum allowable thickness of 2.79 mm, with design limits of 345 kPa(g) and 121 °C. QSCP-RCA-2007-002 determined the failure was caused by localized corrosion resulting from condensing acid gas at the elbow.

Root Cause and Contributing Factors

The investigation identified that corrosion was concentrated at the fitting, which was not monitored. Contributing factors included the placement of CMLs exclusively on straight pipe runs and the reliance on spot readings, which failed to capture the localized thinning at the elbow geometry.

Corporate Requirements

All sites must review inspection layouts to ensure fittings are included in monitoring programs. Paul Granger of Corporate Engineering requires that CMLs be distributed to capture areas susceptible to localized condensation rather than relying solely on straight-run measurements.

Sign-off	Name	Function	Date	Signature
Prepared	Paul Granger	Corporate Engineering, P.Eng.	2008-02-08	_____